

Context-Awareness in Mobile Web Services¹

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Abstract. Context-aware computing is a computing paradigm in which applications can take advantage of contextual information. Quality of network connection is a very important factor for mobile web services. However, the conditions of mobile networks may change frequently and dynamically. Thus, providing support for context-aware applications is especially important in mobile web services. Recently, a number of architectures supporting context-aware applications have been developed, but little attention is paid to the special requirements of mobile devices which particularly have many constraints. This paper discusses a client-proxy-server architecture that supports context-awareness by considering types of device, network and application characteristics. The contribution of this paper mainly lies in the division of labor between proxy and server. Application specific proxy is used to tailor the original resource based on the mobile user's context information. To prove the feasibility, a context-aware image management system is designed and realized.

1 Introduction

With the popularity of mobile devices, people can communicate easily at anytime anywhere. In current market, mobile devices are able to support different platforms. It is necessary to maintain a centralized storage with an easy access interface, so it can provide guaranteed QoS (Quality of Service) to any wired and wireless devices. To achieve this, we have proposed a web-based platform named AnyServer [10], which provides flexible services to different types of mobile (client) devices.

Context is any information that can be used to characterize the situation of an entity. An entity may be a person, place, or object that is considered relevant to the interaction between a user and an application [1]. Context-awareness means that one is able to use context information. A system is context-aware if it can extract, interpret and use context information and adapt its functionalities to the current context. The challenge for such systems lies in the complexity of collecting, representing, processing and using contextual data. The use of context information is especially important in mobile environment, where the quality of network connection provided to mobile users always changes frequently, dramatically, and without warning. Moreover, size and weight constraints on mobile devices limit the computing resources. Besides, battery life is also a nagging concern. As a consequence, mobile applications need to be capable of adapting to these changes to ensure that they could offer the best possible level of service to the user [12], while

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context information is an indispensable component in these applications. In this paper, a client-proxy-server model is built based on our AnyServer platform for using contextual information. Our objectives are (1) to efficiently obtain and transmit contextual information, (2) to use context information to achieve better system performance and user experience, and (3) to implement software architecture for supporting the utilization of contextual information.

The rest of the paper is organized as follows. In Section 2, we introduce some related context-aware applications and their system models. Sections 3 and Section 4 describe our approach to integrate context-awareness in AnyServer platform and the software architecture. The implementation aspects are given in Section 5. We point out future directions and summarize major results in Section 6 and Section 7.

2 Related Work

This section introduces some context-aware mobile applications. The ParcTab system was developed at the Xerox Palo Alto Research Center [2]. The system was developed to experiment with context-awareness in an office environment. The ParcTab worked as a mobile personal digital office assistant. There were also several applications developed for context-aware experimentation. Cyberguide [3] is a computer system that provides information services to a tourist about his/her current location. For example, he/she can find directions, retrieve background information, and leave comments on the interactive map. Knowledge of the user's current location, as well as past locations, is used to provide the kind of service which comes from a real tour guide. More context-aware wireless mobile applications could be found in [6]. Most of the existing applications actually use only few context values, and the most commonly used ones are location, identity and time. The reason for this probably lies in the difficulty for computer systems to obtain and process context information.

Recently a number of architectures were proposed which provided support for context-aware applications in mobile scenarios. Thomas et al. [9] believed that applications had to adapt dynamically and transparently to the resources available at runtime. To achieve this goal, they extended the convention of the client-server model into a client-proxy-server model. Logically, the proxy hid the "mobile" client from the server, which thought that it communicated with a standard client. In [14], the authors identified requirements for a context-aware architecture including light-weight, extensibility and robustness. To meet these requirements, they proposed a three layers architecture which comprised an application layer, a management layer and an adaptor layer. However, all these three layers were located on a single device, so devices exchanged their context information in ad hoc manner and extra requirements were supposed on these mobile devices. Bruce et al. [8] presented a proxy based communication model in which filters executing on an intermediary host drop, delay or transform data moving between mobile device and fixed host. The design and capabilities of this proxy were presented, along with sample filters that addressed "real world" protocols. However, the limitation is that in order to facilitate proxy filtering, all traffic traveling to and from a particular mobile host must pass through a single gateway, the Proxy Server.

3 Context-Awareness in Any Server Platform

To make AnyServer Platform (as shown in Fig. 1) provide adaptive services based on user's context information, some additional sensors or programs are generally required to capture context information. Moreover, a common representation format for such information should be adopted, so different applications are able to use the same context information. Furthermore, AnyServer must intelligently process these information and deduce the meaning. Device-aware, network-aware and application-aware, these three factors are the most useful in AnyServer platform. The contextual data used in AnyServer include user device information (e.g. screen size, computational power and battery life), network conditions (e.g. network connectivity and transmission delay), and application type (e.g. real-time video/audio, or web browsing). Direct context information sharing is not needed in AnyServer platform.

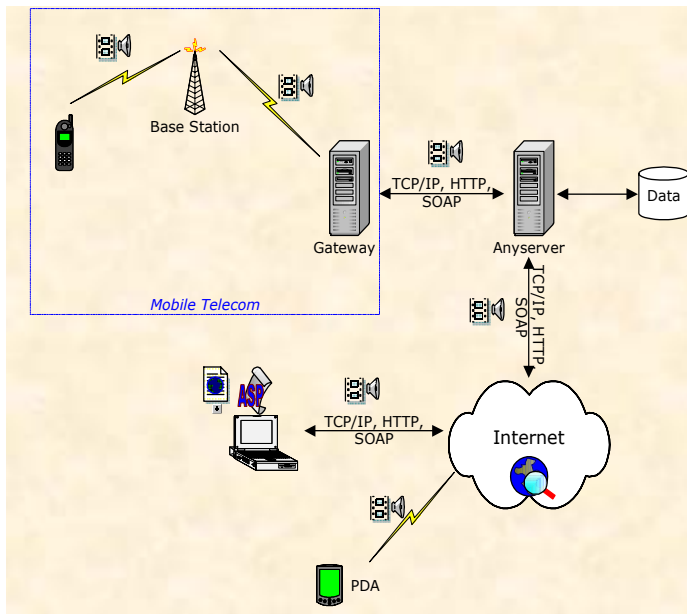


Fig. 1. Any Server Platform

3.1 Information Capturing

In this subsection we will answer the following three How-to questions:

How to Obtain Device Information? Device information retrieval can be based on Universal Plug and Play (UPnP). As depicted in [4], UPnP is designed to support zero-configuration, “invisible” networking, and automatic discovery for a breadth of device categories from a wide range of vendors. In UPnP the properties associated

with the device are captured in an XML based device description document that the device would host or could be obtained from a specific URL. That is when the client sends request to AnyServer, a device description document or a specific URL accompanies with the request. The device information could be managed by AnyServer as part of user profile (to be detailed later) for future use. Based on the device information, some adaptive transmission could be carried out.

How to Capture Network Condition Information? In Any Server platform, round trip time (RTT) is chosen to measure the transmission delay. When measuring RTT several probing messages will be sent out and the following notations will be used to calculate more exact values:

RTT: the record stores the round trip time value of every probing.

EA_RTT: the record stores the exponential average of *RTT*.

Considering dynamic estimate of variability in estimating RTT, *EA_RTT* is averaged as:

$$EA_RTT(n) = \beta * EA_RTT(n-1) + (1 - \beta) * RTT(n) \quad (1)$$

We would like to give the greater weight to more recent instance of RTT. Thus we use $\beta = 0.1$. If the mobile device has strong computational capability, it is responsible to conduct the probing. Otherwise the probing is the responsibility of the proxy. Network condition is runtime information and will be captured every now and then.

How to Know Application Related Information? Applications supported by AnyServer platform can be divided into two kinds based on timing consideration and tolerance of data loss. Timing consideration is important for many applications that are highly delay-sensitive but loss-tolerant, such as real-time interactive audio and video. Meanwhile, tolerance of data loss is important for many applications that are not loss-tolerant but delay non-sensitive, such as file downloading. In AnyServer platform, different applications will be served by different proxies.

3.2 Information Representing

Instead of creating a new kind of description language, we use eXtensible Markup Language (XML) for representing contextual information. XML is the universal format for structured data on the Web. And XML can place nearly any kind of structured data into a text file. Also the platform independent feature of XML easily tackles the obstacle that the heterogeneity of mobile devices brings to us. And using XML we can easily extend our contextual information to extra aspects. When a new family of mobile device is evolving, we can reuse the design for the old devices in an optimal way. As a result, XML is used in contextual information description. Fig. 2 is an example of device information depicted in XML.

4 Software Architecture

AnyServer must have the ability to change its behavior depending on the current context of users. In this way, a lot of personalized applications can be built based on

```
<?xml version="1.0" encoding="big5"?>
  <Device>
    <Resolution>
      <Properties>
        <Property NAME = "Height"> 800 </Property>
        <Property NAME = "Width"> 600 </Property>
      </Properties>
    </Resolution>
    <Screen>
      ... ..
    </Screen>
    ... ..
  </Device>
```

Fig. 2. XML example of device information

Any Server platform. Some architectures [8][9][14] have been introduced in Section 2. Our approach is the extension of the conventional client-server model, which is also used in [9] but with quite different functionalities. In [9], the proxy shares the application logic of the “standard client” with mobile clients to adapt to dynamic wireless environment and to address the limitations of portable devices. But in our architecture the proxy relatively shares the logic of “standard server” with AnyServer. More details will be given below.

4.1 The Approach in Any Server Platform

Our client-proxy-server model naturally comes from the requirement of system scalability. To alleviate the load of AnyServer and provide the best level of service to mobile users, a proxy is introduced to the traditional client-server model. This model is primarily made up of three components listed below:

- Client: The mobile client can be PDA (Personal Digital Assistant), mobile phone or laptop. It can enjoy the attractive applications provided by AnyServer platform.
- Proxy: The user traffic would be redirected to the proxy if necessary. The proxy could also be used to probe the network condition of mobile users and tailor the original data for adaptive transmission.
- AnyServer: The server provides intelligent integrated services, such as short message services, image uploading and downloading, on-line audio and video. It is also the context information server that manages these information as user profiles.
- This communication among client, proxy and AnyServer is shown in Fig. 3. The detail is depicted as follows:

1. The client starts by transmitting the request to AnyServer.
2. Based on the application type, AnyServer responds to the client or redirects the user traffic to the application specific proxy.
3. AnyServer sends the original content and the related user profile to the proxy.
4. The client sends the same request to the proxy, maybe the network condition information gathered by the client is also sent to the proxy.
5. The proxy responds to the client with the tailored context based on the user's context information.

For the limited computational resource of mobile clients, an alternative way is that the proxy actively probes the network condition of the client. In other words, step 4 and step 5 could be replaced by unidirection traffic. However, it is also a tradeoff between the load of the client and the proxy. Moreover, the introduction of proxy can also tackle the puzzle of conflicts among adaptations of different independent applications, while this conflict is a common potential problem that can occur in a system utilizing separate adaptation mechanisms for different applications. In our approach, each proxy is designated to serve a specific application.

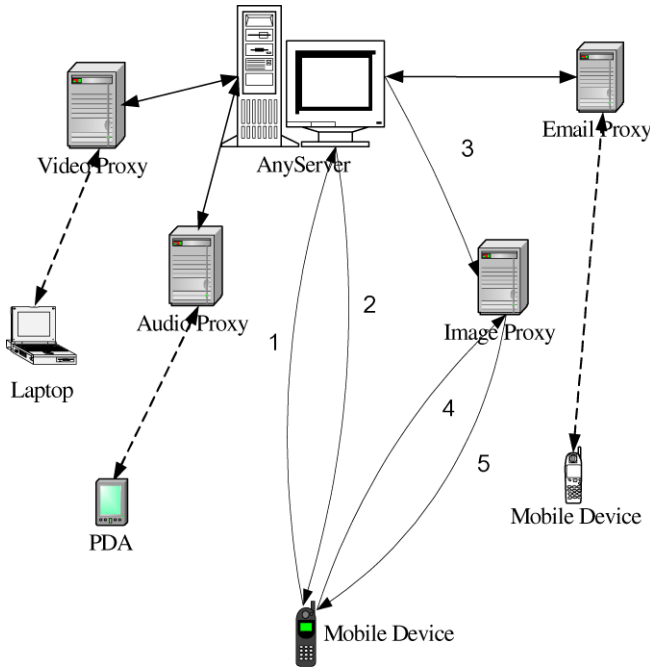


Fig. 3. Communication among client, proxy and AnyServer

4.2 User Profile Management

Customizations of mobile services to various kinds of terminals, user preferences, and varying network characteristics are attractive aspects in mobile computing. Different applications typically have different preferences for latency and integrity of data. Mobile clients need to express their preferences using voluntary profiles. The simplest way to manage the user profiles is to use a centralized context server, which provides contextual information to the applications. Schilit's mobile application customization system [5] contained dynamic environment servers, which managed a set of variable names and values representing an environment. It delivered updates to clients that had previously shown interest by subscribing to the server. The Rome system [7] developed at Stanford was based on the concept of a context trigger, which consisted of a condition and an action. This system allowed decentralized evaluation of triggers by embedding triggers in end devices. However, it did not allow context sharing.

In our architecture, user profile can be managed by AnyServer platform. Examples of user profiles are: (1) User Information Profile, such as name, identity, e-mail address; (2) Device Profile, such as device type, screen capability, memory size; (3) User Preference Profile, such as the preferred display pattern, favorite background color. User profile can also be used to save the session information. Due to the poor wireless network connection, mobile users may connect to AnyServer many times to complete one task. If the related information of each session can be properly saved, the remainder task of the previous session can be resumed in the new session when the mobile users reconnect to AnyServer platform.

5 Implementation Aspects

To prove the feasibility of our presented architecture, a prototype has been developed on AnyServer platform. Pocket PC is chosen as the first client device in the first phase. Pocket PC Client (PPCClient) is developed with Microsoft .NET Compact Framework. The reasons of this combination can be concluded as follows:

1. If a GSM (Global System for Mobile) modem is inserted into a Pocket PC, the Pocket PC will be a PDA phone that can connect to the Internet.
2. The popularity of smart phones with Microsoft .NET Compact Framework is gradually increasing, so that the programs developed for Pocket PC can be easily migrated to those smart phones.
3. More functionalities can be implemented for Pocket PC than mobile phones currently due to lack of processing power and computing capability of the mobile phones.

The PPCClient is running on Pocket PC 2003 operating system, which is the same as that for smart phones. Using Pocket PC, we can choose either IEEE 802.11 or GPRS (General Packet Radio Services) to connect to Internet. Besides, C# and embedded C++ are used as programming languages.

In Section 5.1, AnyServer, the proxy and mobile client are introduced. Uploading and downloading image are used as examples for context-awareness implementation in AnyServer platform, and they are presented in Section 5.2.

5.1 Any Server, Proxy and Mobile Client

The server side in AnyServer platform, called AnyServer, is a physical server linked with a database. For the mobile clients, the request can be sent through different network protocols in order to provide flexibility and adaptability features to those heterogeneous client devices that may only accept one specific network protocol. Database is a vital component in AnyServer platform, the user profile is managed by the database. General speaking, the proxy has two main functionalities: one is to gather the network condition information; the other is to tailor the original resource provided by AnyServer based on the mobile user's context information. In Fig. 4, the proxy may take following actions: (1) Degrade video stream, that is selectively dropping some structured data, such as frames in MPEG data stream; (2) Drop heuristically some unstructured data, such as quoted text from email body; (3) Compress data instead of dropping it. These actions are application specific. For different applications, the proxy would take different filter policies.

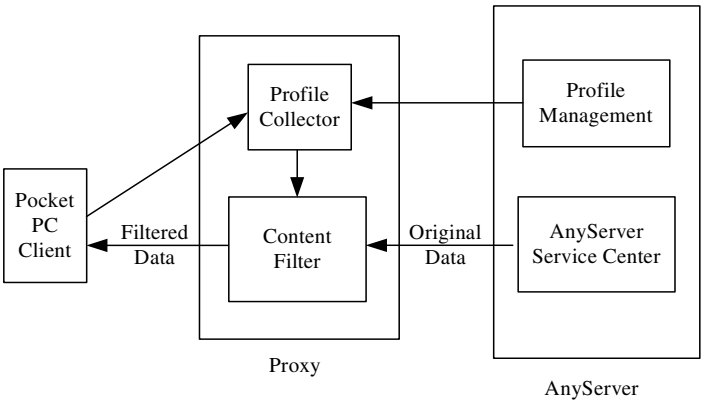


Fig. 4. Data Flow in Any Server platform

5.2 Example: Context-Aware Image Management System

In this section, an image management system is implemented based on AnyServer platform. When a user retrieves his multimedia materials on AnyServer platform, it involves two actions. Firstly, the user will get all the headers of the multimedia materials belong to him/her. As a result, the corresponding XML format headers information will be retrieved and therefore it reduces the network transmission volume. This feature will benefit those mobile users who use the service through mobile telecommunication networks which charge the user based on the volume of data sent. Then, when the user really selects a particular item to view, the actual content (maybe filtered) will be downloaded from the server side to the client device. Fig. 5 is an example of image filtering and shows both the original image and the filtered one.



Fig. 5. Image Filtering

6 Future Works

Context-aware computing has been proved to provide a reliable and flexible platform, especially for resource limited mobile devices. A natural thinking is to extend this platform to support end-to-end communications between terminals. Our future work is to build an end-to-end multimedia communication system. In this system, adaptive transmission technology will be realized and the proxy will support more functionalities.

Adaptive transmission is a hot research topic in wireless networking [11][12] where various mobile devices must be supported and the network condition fluctuates dynamically. The relative resource poverty of mobile elements as well as their lower trust and robustness argues for reliance on static servers. But the need to cope with unreliable and low-performance networks, as well as the need to be sensitive to power consumption argues for self-reliance [13]. Any viable approach to mobile computing must strike a balance between these competing concerns. As the circumstances of a mobile client change, it must react and dynamically reassign the responsibilities of the proxy. In other words, the proxy must be adaptive.

With the development of mobile devices, users may desire not only communicating with servers, but also direct end-to-end data transmission. However, it is not easy for various types of wireless devices of different capabilities to communicate with each other. For example, a mobile phone could not play the video file of some format transferred from a laptop. We are now building a system to support reliable, comfortable and compatible end-to-end communication. At the moment, the role of proxy in our AnyServer platform is just a content filter, and it tailors original content based on the context information of mobile users. In our future work, the proxy can evolve into a content server and AnyServer will take the role of an index server. Under this framework, more challenges will be faced, such as distributed content management and update, resource allocation between content servers.

7 Conclusion

Context-awareness offers many new possibilities for mobile web services. Motivated by this, we propose in this paper a client-proxy-server architecture trimmed to the special needs of mobile users. This architecture takes the special requirements of the mobile devices, such as limited capability of memory and power, into consideration. Moreover, it supports context-awareness by considering types of device, network and application. This architecture brings division of labor between proxy and server. Application specific proxy provided by AnyServer platform is used to tailor the original resource based on the mobile user's context information. We also implement our architecture and demonstrate its feasibility by a context-aware application.

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